

# tf.compat.v1.orthogonal\_initializer Stay organized with collections Save and categorize content based on your preferences.

[https://www.tensorflow.org/api\\_docs/python/tf/compat/v1/orthogonal\\_initializer](https://www.tensorflow.org/api_docs/python/tf/compat/v1/orthogonal_initializer)

*Initializer that generates an orthogonal matrix.*

## Function Signatures

```
tf.compat.v1.orthogonal_initializer( gain=1.0, seed=None,  
dtype=tf.dtypes.float32 )  
(shape[0] * ... * shape[n - 2], shape[n - 1])
```

## Code Examples

```
tf.compat.v1.orthogonal_initializer(  
gain=1.0,  
seed=None,  
dtype=tf.dtypes.float32  
)
```

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tf.compat.v1.orthogonal_initializer(  
    gain=1.0,  
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)
```

```
tf.dtypes.float32
```

```
@classmethod  
from_config(  
    config  
)
```

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@classmethod
```

```
from_config(  
    config  
)
```

```
initializer = RandomUniform(-1, 1)  
config = initializer.get_config()  
initializer = RandomUniform.from_config(config)
```

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initializer = RandomUniform(-1, 1)  
config = initializer.get_config()  
initializer = RandomUniform.from_config(config)
```

## Documentation

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Initializer that generates an orthogonal matrix.

## View aliases

Compat aliases for migration

See

Migration guide for  
more details.

`tf.compat.v1.initializers.orthogonal`

If the shape of the tensor to initialize is two-dimensional, it is initialized with an orthogonal matrix obtained from the QR decomposition of a matrix of random numbers drawn from a normal distribution.

If the matrix has fewer rows than columns then the output will have orthogonal rows. Otherwise, the output will have orthogonal columns.

If the shape of the tensor to initialize is more than two-dimensional, a matrix of shape  $(\text{shape}[0] * \dots * \text{shape}[n - 2], \text{shape}[n - 1])$  is initialized, where  $n$  is the length of the shape vector.

The matrix is subsequently reshaped to give a tensor of the desired shape.

## References

`## Methods`

### `from_config`

[View source](#)

Instantiates an initializer from a configuration dictionary.

### Example:

`## get_config`

[View source](#)

Returns the configuration of the initializer as a JSON-serializable dict.

`__call__`

[View source](#)

Returns a tensor object initialized as specified by the initializer.